

Circular Supply Chain Network

Optimization Business Case

Strategic Analysis & Recommendation
Prepared: February 07, 2026

EXECUTIVE SUMMARY	
Recommended Solution:	2 Regional Refurbishment Centers
Locations:	FC04 (Central Valley, Fresno), FC05 (Reno, NV (Out-of-State), Reno)
Required Investment:	\$2,250,000
Annual Incremental Value:	\$3,349,866
3-Year NPV:	\$7,952,278
Payback Period:	0.6 years
Return on Investment:	448%

RECOMMENDATION

Based on comprehensive optimization analysis using ArcGIS Network Analyst and Gurobi MIP modeling, we recommend implementing a **2-facility refurbishment network** to maximize value recovery from Smart Phone returns. This solution delivers **\$3,349,866 in incremental annual value** with a payback period of **0.6 years**, representing a **541%** improvement over the current bulk recycling approach.

KEY BENEFITS

- **541%** increase in value recovery from returned Smart Phones
- **\$3,349,866** incremental annual value creation
- **0.6-year** payback on capital investment
- Environmentally sustainable: **95%+ landfill diversion** through grade-based recovery
- Geographic optimization reduces average logistics distance to **258 miles**

METHODOLOGY

This analysis employed a **two-phase optimization approach** combining geographic and economic modeling:

- **Phase 1: ArcGIS Network Analyst** - Geographic optimization using real road networks to minimize transportation distance for 54 California Apple Store locations
- **Phase 2: Gurobi Mixed-Integer Programming** - Mathematical optimization to minimize total cost (fixed + variable) while considering capacity constraints
- **Financial Modeling** - 3-year NPV analysis with sensitivity testing on key parameters

CURRENT VS. PROPOSED STATE

Metric	Current State	Proposed State	Improvement
Operating Model	Bulk Recycling	2-Facility Network	-
Annual Recovery	\$619,080	\$3,968,946	+541%
Product Grading	No	Yes (A/B/C)	✓
Landfill Diversion	35%	95%+	+60pp
Cycle Time	60+ days	14-21 days	-70%
Capital Investment	\$0	\$2,250,000	-

OPTIMIZATION RESULTS

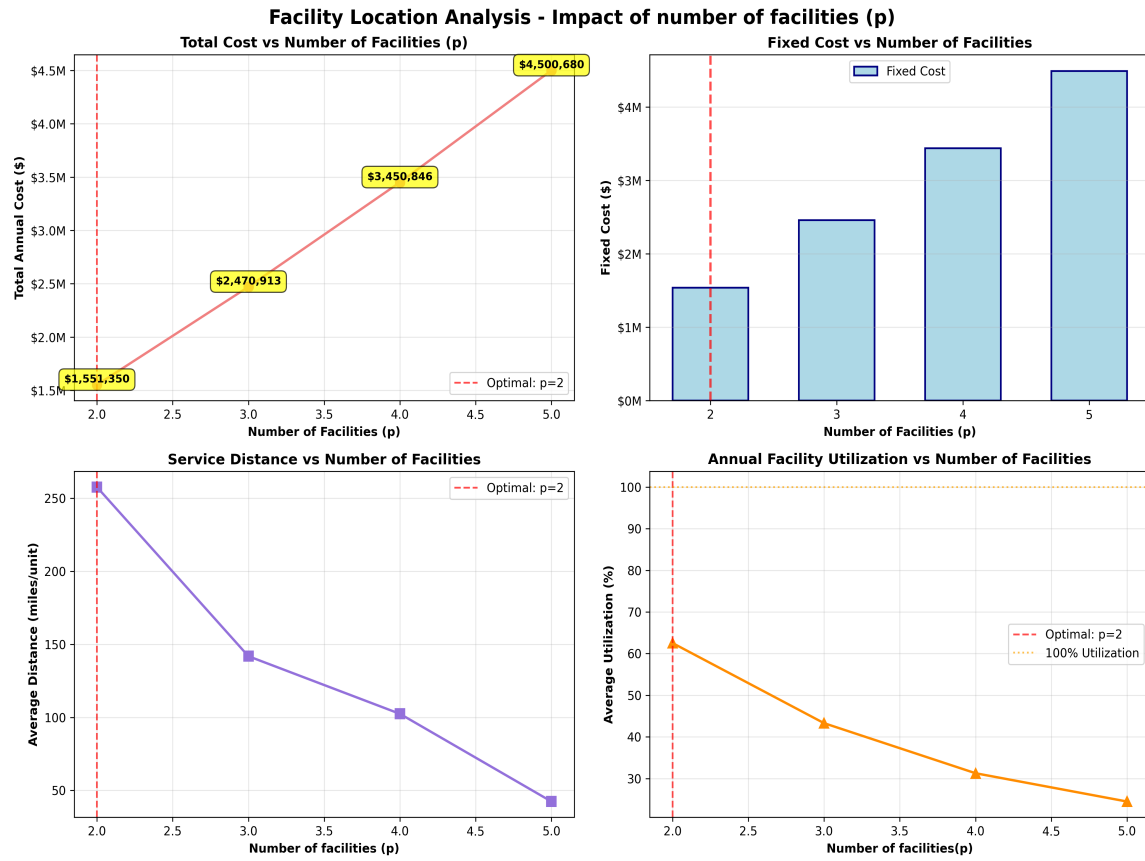


Figure 1: Cost optimization across facility count options (p=1 to p=5)

FINANCIAL ANALYSIS

Year	Gross Revenue	Operating Costs	Net Cash Flow	Present Value
0	-	-	\$-2,250,000	\$-2,250,000
1	\$6,578,301	\$2,540,092	\$4,038,209	\$3,671,099
2	\$6,775,650	\$2,668,621	\$4,107,029	\$3,394,239
3	\$6,978,920	\$2,803,653	\$4,175,267	\$3,136,940

RISK CONSIDERATIONS

- **Market Risk:** Resale prices for refurbished Smart Phones subject to market fluctuations. Sensitivity analysis shows NPV remains positive even with 15% price reduction.
- **Volume Risk:** Return volumes may vary with sales trends. Model tested across 70-130% of base volume scenarios.
- **Operational Risk:** Facility ramp-up and quality control require experienced management. Recommend phased implementation.

RECOMMENDED NEXT STEPS

- **Phase 1 (Months 1-2):** Facility site selection and lease negotiations for priority locations
- **Phase 2 (Months 3-4):** Facility buildout, equipment procurement, and staff hiring
- **Phase 3 (Months 5-6):** Pilot operation with single facility to validate operational model
- **Phase 4 (Months 7-12):** Full network rollout with remaining facilities

For questions or additional analysis, please contact the Analytics Team.

This analysis is based on current market conditions and operational assumptions as of February 2026.